

The background is a deep blue gradient with faint, stylized celestial patterns. On the left, there are concentric circular arcs and a semi-circular scale with numerical markings from 140 to 260. Scattered throughout are smaller circular motifs, some with dashed lines and arrows, suggesting orbital paths or astronomical instruments.

70 YEAR HISTORY OF THE PHOENIX ASTRONOMICAL SOCIETY

NOVEMBER 1948 TO OCTOBER 2018

1ST MEETING NOV 15, 1948

12 Charter Members including

Dr. Bill Rhodes PhD President

Professor Amos Hoff Director and Sponsor

Members came from Phoenix College

Arizona State and elsewhere



DOC RHODES' 16 INCH HANDMADE
GREGORIAN TELESCOPE



24 INCH CLUB TELESCOPE MIRROR COMPLETED 1954



24 INCH COMPLETED TELESCOPE



PAS OBSERVATORY AT
PHOENIX COLLEGE



1964 LAGOON NEBULA

40 MIN EXPOSURE



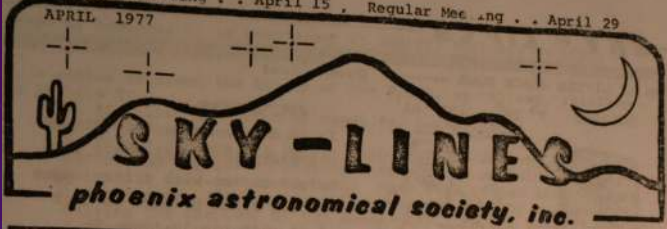
1964 M15 GLOBULAR

30 MIN EXPOSURE



1977 2 PAGE PAS NEWSLETTER

Special Meeting . . April 15 . Regular Meeting . . April 29
APRIL 1977



SKY-LINES
phoenix astronomical society, inc.

TWO MEETINGS, FRIDAY 15TH & 29TH

This month there will be two meetings, both on a Friday, and both at Phoenix College, room C102 at 8:00 PM. The first meeting will be a special meeting requested, as per the bylaws, by six club members. This special meeting, on April 15 will be involved with some suggested changes in the bylaws. Complete details are included elsewhere in this newsletter.

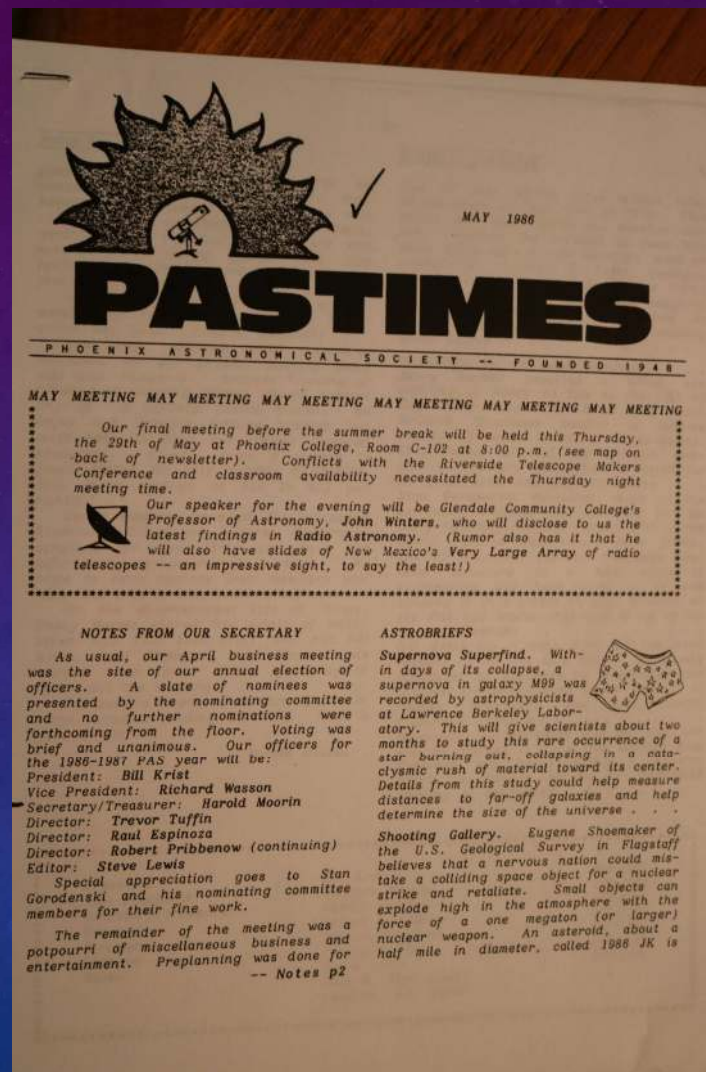
The second meeting, on April 29, will be the yearly business meeting at which time the officers and directors for the next year will be elected. Also to be discussed will be the fate of the 2 1/2 inch scope. Please plan to attend both meetings as a number of very important items will be examined.

Another reminder is in order here also. The Phoenix Astronomical society, in cooperation with Phoenix College will be offering this summer a course called Astronomy for Hobbyists. This course will be for people who have not had any previous experience with scopes and could just as well be called "Everything (almost) that you've ever wanted to know about telescopes but didn't know who to ask". Dates are from June 7 to July 5 inclusive from 7:00 to 10:00 PM each Tuesday evening in room E112. If you happen to know of some people who might derive some benefit from this course, please let them know. Additional details will be coming out shortly, but additional information can be obtained immediately by calling Phoenix College, 264-2192 and asking for either Dean Waldron or myself. Hopefully, if this course is a success, it can be offered on a regular basis in an effort to inform the public about this hobby.

Hope to see you. *Bill Anderson*

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President - Ken Sikes, Secretary-Treasurer - Chuck Nash
Opinions expressed are those of the writers and not necessarily that
of the Phoenix Astronomical Society or its officers.
EDITOR : Cliff Vaughan 838-7107, Assoc. Editor : Florence Bossee

1986 3 PAGE PAS NEWSLETTER





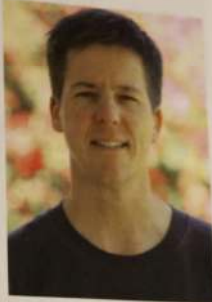
PAStimes

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October 2017
Volume 69 Issue 2

PHOENIX ASTRONOMICAL SOCIETY — ESTABLISHED 1948

Seeking Supernovae in Gravitational Waves

Provided by Kai Staats

Michele Zamboni

Kai Staats

September 14, 2015, the Laser Interferometer Gravitational-wave Observatory (LIGO) received a most auspicious signal was later confirmed to be from a pair of black holes over 1 billion light years from Earth. On the 100th anniversary of Einstein's theory of relativity, this event marked the first detection of black holes and the end of nearly 40 years of work by physicists world-wide. Now, LIGO concluded its second official observation "O2", with more detections and future unfolding.

Arizonans realize that a leading search team is in our own backyards. Michele Zamboni (mee-khel-lay) and his research team, including Kai, are looking at the signal of a supernova in gravitational-wave data, a challenging task. (Continued on page 3)

PAS Meeting of Sept 7

By Terri, Event Manager

Insana opened the meeting at 7pm with the installation of the new officers of the club. Mike Marron took over as President and Sam became Past President. On page 3, there is an article about the new officers of PAS for the year.

Then turned the meeting over to Howard who presented on the Solar Eclipse of August 21, they were:

- Howard told us about his trip to Oregon with his son Frank, Mike and Ofelia.
- Paul Facuna, who was at that same location as Paul Maley.
- Ron Walker who provided some great photos of his experience.
- Don Boyd who shared what William

(Continued on 3)



At the September PAS Meeting, Howard asked William to take a photo with his phone. Howard wanted to have the three shirts as one awesome photo of Howard wearing the Curiosity Rover, Sam has the Total Solar Eclipse, and Terri is wearing (all black) Totality. Photo provided by Howard Moneta.

CLYDE TOMBAUGH 1986 PAS MEETING



PAS STAR PARTY

telescopes



binoculars



PAS STAR PARTY LOCATIONS OVER THE YEARS

- *Phoenix College
- *Castle Hot Springs
- *Table Mesa Road
- *McDowell Mountain Park
- *Lake Pleasant
- *Thunderbird Park
- *New River
- *Black Canyon City
- *Antennas
- *Roadrunner Park
- *Mayer
- *Bookmans
- *Mike's Carefree
- *Az Science Center
- *PVCC BMC
- *South Mountain Park
- *Tonto National Monument

METEORITE DISPLAYS



PAS FIELD TRIPS

Kitt peak observatories

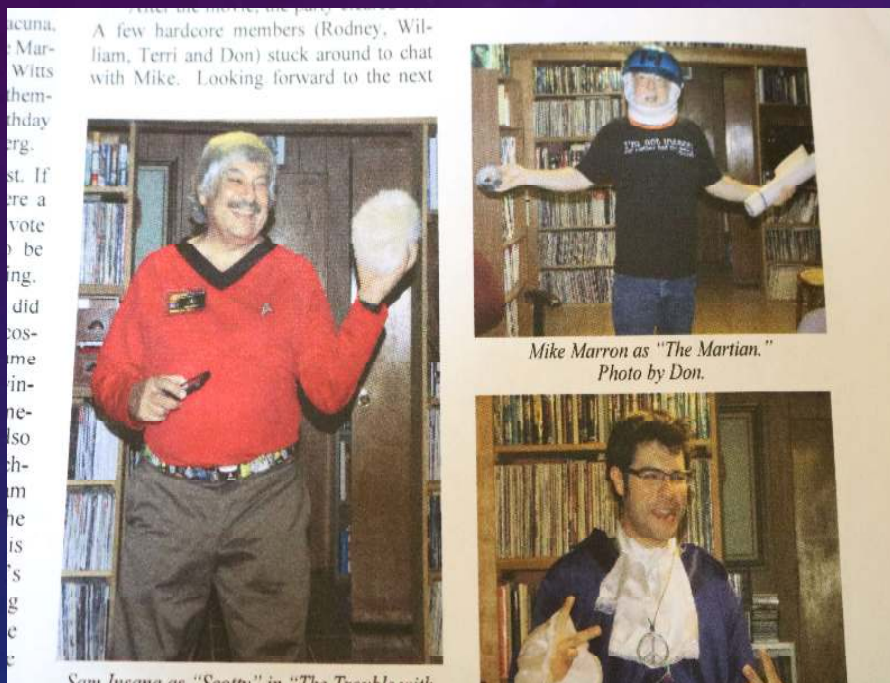


Meteor crater



PAS SOCIAL ACTIVITIES

Halloween Costume Party



Music Jam



HAZARDS OF DARK SKY VIEWING IN THE DESERT



PAS TELESCOPE WORKSHOP



METEOR SHOWER CITIZEN SCIENCE PROJECT

International Meteor Organization

Observational data sheet



International Meteor Organization

Home Organization Meteor Science Meteor Observations

Home » Meteor Science » Observation Methods » Visual Meteor Observation » Major Showers Observations » The observation

When a meteor appears

Other languages: [Español](#)

Having read to this point you may have got the impression that meteor observing consists mainly of recording lots of odd data unconnected with meteors. But that is not the case. Now we reach our main subject: What do you have to do when a meteor appears? The most important thing is to remain calm so that you can record what you have just seen as objectively as possible. Keep looking at the region of sky where you saw the meteor in order to memorize the following data in this order:

Meteor's path

The most important information is its direction rather than the exact length and start/end points. Memorize it with respect to the background stars.

Organization

- Home
- Introduction
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- Membership

Meteor Science

- Observations
 - Visual
 - Minor Showers
 - Major Showers
 - When
 - Where
 - Equipment
 - Preparations
 - The observation
 - What to record
 - Limiting
 - Magnitude

Visual meteor observing report

```
// Header section
observer: Sam, Insana
site: Carefree, Arizona, United States (-111.908 E, 33.833 N)

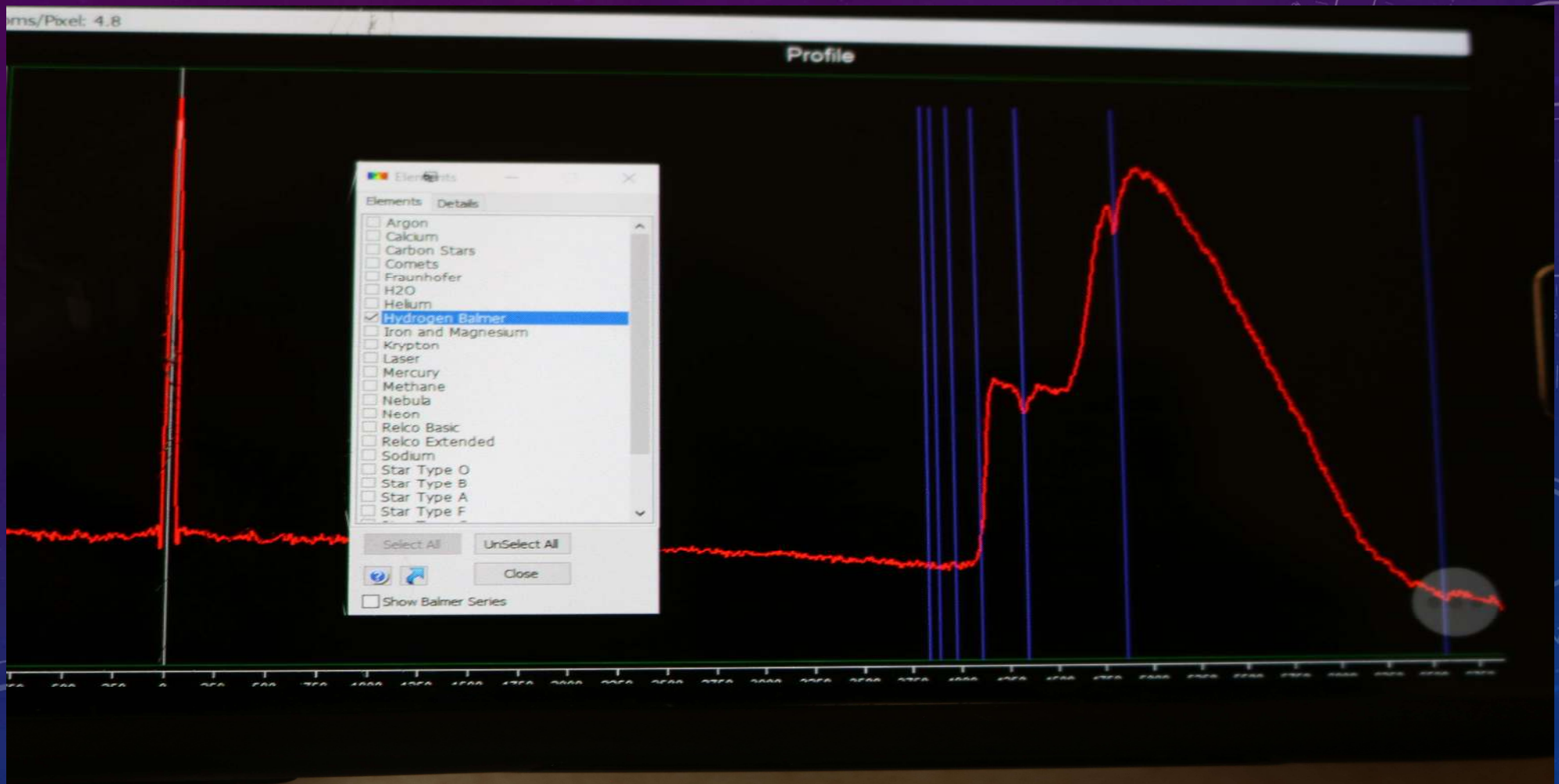
// Number section
```

	start	stop	RA	DEC	Teff	F
period	2015-12-14T04:50:00	2015-12-14T05:20:00	105	20	0.500	1.00
period	2015-12-14T05:48:00	2015-12-14T06:03:00	105	20	0.250	1.00
period	2015-12-14T06:03:00	2015-12-14T06:18:00	105	20	0.250	1.00
period	2015-12-14T06:45:00	2015-12-14T07:00:00	105	20	0.250	1.00
period	2015-12-14T07:00:00	2015-12-14T07:15:00	105	20	0.250	1.00

```
// Magnitude section
```

	Shower	start	stop	-6	-5	-4	-3	-2
distribution	GEM	2015-12-14T04:50:00	2015-12-14T05:20:00	0	0	0	0	0
distribution	SPO	2015-12-14T04:50:00	2015-12-14T05:20:00	0	0	0	0	0
distribution	GEM	2015-12-14T05:48:00	2015-12-14T06:03:00	0	0	0	1	0
distribution	GEM	2015-12-14T06:03:00	2015-12-14T06:18:00	0	0	0	0	1
distribution	GEM	2015-12-14T06:45:00	2015-12-14T07:00:00	0	0	0	0	0

SPECTROSCOPY

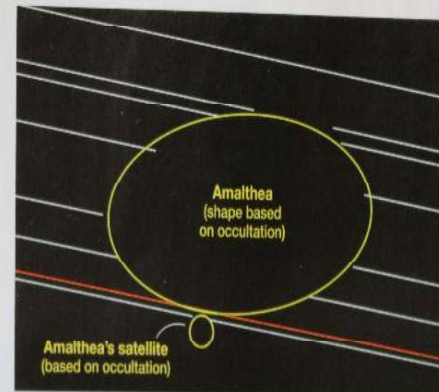


OCCULTATION TIMING CITIZEN SCIENCE PROJECT



NEWS NOTES

PRO-AM Amateurs Find Asteroid's Moon



▲ These lines represent the projected paths of a 10th-magnitude star recorded by observers on March 14, 2017, as the star passed behind the asteroid Amalthea. Sam Insana in Gila Bend, Arizona, measured the red track, and the white line below it is actually two tracks close together. The yellow oval marks the asteroid's approximate shape; the small yellow circle just below it corresponds to the location of the potential moon.

A TEAM OF AMATEUR OBSERVERS, some armed with just 3-inch telescopes, found that the main-belt asteroid 113 Amalthea probably has a small companion.

Paul Maley, a retired NASA staffer and a key member of the

the definitive compilation by W. Robert Johnston, the census of asteroids with moons now includes 145 main-belt asteroids (8 with two each), 23 Mars-crossing asteroids (1 with two), and 63 near-Earth asteroids (2 with two each). Looking farther outward, Johnston's list includes 4 of Jupiter's Trojans and 82 trans-Neptunian objects.

Nor is it clear what kind of orbit the newfound companion might have. According to Daniel Green (Harvard University), small satellites around large main-belt asteroids typically have orbits roughly five times larger than their primary's diameter, which here would correspond to about 250 km. But the observed gap was far smaller, so perhaps the companion's orbital plane was seen nearly edge-on with the satellite near one of its conjunction points (rather than being far off to one side).

The whole claim to a satellite discovery rests on Insana's negative observation — take that away, and the two brief outages recorded by Eisefeldt and Campbell could simply be due to a large bump on Amalthea's surface. However, Maley says, "The Insana video has been checked over and over, as was his location, and the only conclusion that we can come to is that the miss is completely clean."

Confirmation of the discovery might not have to wait long. Although no one had previously observed a stellar

OTHER ASTRONOMY ACTIVITIES AND CITIZEN SCIENCE PROJECTS PAS MEMBERS HAVE ENJOYED

- ASTROPHOTOGRAPHY
- THE GREAT WORLD WIDE STAR COUNT
- HERSCHEL 400 OBSERVING PROGRAM
- HERSCHEL II OBSERVING PROGRAM
- MESSIER OBSERVING PROGRAM
- DOUBLE STAR OBSERVING PROGRAM
- DARK SKY ADVOCACY
- PUBLIC OUTREACH OBSERVING AWARD
- OTHER ASTRONOMICAL LEAGUE OBSERVING PROGRAMS

PASAZ.ORG PHOENIX ASTRONOMICAL SOCIETY WEBSITE

Phoenix Astronomical Society Website Screenshot

Established 1948

Phoenix Astronomical Society

Phoenix Astronomical Society

User Name Password

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Welcome to PAS



Phoenix Astronomical Society
Established 1948
Located in Phoenix Arizona

Let P.A.S. be your PORTAL TO THE STARS!

We have many science and space related interests. We are amateur astronomers with a desire to learn more about the night sky and to share this knowledge with others. Welcome! But wish to further this knowledge. Please email us to learn more about PAS.

» Current Moon

CURRENT MOON



Waxing Crescent
37% of Full
Thu 18 Aug. 2016
[Lunar Phases](#)

» Upcoming Events

08-19-2018
» Free Telescope Workshop at Bookmans

09-06-2018
» Dinner before the Meeting: 4:30pm to 6:30pm
» PAS General Meeting with Guest in Room LS 203

August 2016

S	M	T	W	T	F	S
29	30	31	1	2	3	4

Type here to search

Phoenix Astronomi...

PAS HOLIDAY SOCIAL